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[Measuring Outcomes and Using]

› KPI

[Product Management: Designing and Running]

› Experiments

KPI's

SMART

SMART –

Question: Which factors should you consider when choosing a performance measure?

Instruction: Choose all options that best answer the question.

- The popularity of the measure in your industry
- A balanced focus on both the short and long run
- A focus on key success factors
- The degree of hierarchy in the organization
- A focus on performance that managers can control or influence
- Whether the measure is easy to understand
- The promotion of goal congruence

Question: What are the key advantages of using a performance measurement system?

Instruction: Choose all options that best answer the question.

- It reduces managers' workloads
- It enables you to identify and assess triggers for changes in performance
- It guarantees performance improvement for all employees
- It builds a connection between strategy and daily operations
- It makes it easier to communicate high-level strategic vision
- It makes it possible to manage performance proactively

Question: What are the attributes of good KPIs?

Instruction: Choose all options that best answer the question.

- They evaluate success in reaching targets
- They are a form of communication
- They incorporate feedback from managers, analysts, and department heads

They involve collecting very high volumes of data
They involve measurable values
They are broad and nonspecific

Question: What are some ways you can define and clarify the criteria for performance measures?

Instruction: Choose all options that best answer the question.

Use criteria established by other companies
Distinguish between company and business unit level measures
Define the elements of each performance measure
Establish incentive and reward systems
Have managers rate the importance of different company goals

Question: Place the steps for selecting appropriate KPIs in order.

Result: Correct. Great job!

1. Define the organization's goals
2. Define the organization's objectives
3. Determine how to reach the objectives
4. Choose KPIs

Question 1 of 5

Question: What are the advantages of using a performance measurement system?

Instruction: Choose all options that best answer the question.

It can help reduce an organization's overhead costs
It builds a link between day-to-day operations and strategy
It facilitates communication of top-level strategic visions
It enables an organization to keep track of all minor metrics
It makes it possible to examine triggers for changes in performance
It puts you in a better position to manage performance proactively

Question 2 of 5

Question: What are the key attributes of good KPIs?

Instruction: Choose all options that best answer the question.

They evaluate success in reaching targets
They compare performance to other companies
They are a form of communication

- They focus purely on financial measures
- They incorporate feedback from analysts, department heads, and managers
- They involve measurable values

Question 3 of 5

Question: Which factors should you consider when choosing a performance measure?

Instruction: Choose all options that best answer the question.

- Association with a goal that's readily achievable
- Focus on what managers can control or influence
- Use of in-depth statistical tools
- The promotion of goal congruence
- How easy it is to understand the measure
- Focus on key success factors
- A balanced focus on both the short and long term

Question 4 of 5

Question: To select suitable KPIs, you should follow a particular sequence of steps.

Place the steps in the proper order.

Instruction: Rank the following items in the correct sequence, by dragging or using the move up and down buttons.

1. Define your organization's strategic goals
2. Define your organization's objectives
3. Determine how to reach your objectives
4. Choose KPIs

Question 5 of 5

Question: How can you define and clarify criteria for performance measures?

Instruction: Choose all options that best answer the question.

- Select relevant KPIs
- Establish incentives and reward systems
- Use trial and error to get the desired results
- Define the elements of the performance measure
- Distinguish between measures at the company and business unit levels

Product Management: Designing and Running Experiments

Job Aid: How to Run a Business Experiment

Conducting business experiments can have a huge payoff, but they require a huge investment. Before launching an experiment, you need to ask some key questions to ensure the experiment will be worth that investment.

Question

Considerations

Does the experiment have a clear purpose?

- Only conduct experiments if they are the only practical way to answer specific questions about product development issues
- First figure out exactly what you want to learn, then decide if testing is the best approach
- Determine the scope of the experiment
- Investigate the ancillary effects of an experiment, both positive and negative

Have stakeholders made a commitment to abide by the results?

- Ensure stakeholders agree how they'll proceed once the results are in
- Weigh all the findings instead of picking data that supports a particular point of view
- Put a process in place to ensure results aren't ignored
- Ensure experiment support the organization's priorities
- Be willing to walk away from a project if results show it's not worth pursuing

Is the experiment doable?

- Remember the complexities of variables and their interactions can make it difficult to determine cause-and-effect relationships
- Remember that large enough sample sizes can mitigate effects of complex variables
- Use a smaller sample size if you expect a cause to have a large effect; use a larger sample size if you expect a small effect
- Select the right sample size to
 - Ensure results will be statistically valid
 - Enable the company to decrease testing costs
 - Enable the company to increase innovation

Can you ensure reliable results?

- Use randomized field trials – randomly divide test subjects with the same characteristics into a test group and a control (untested) group
- Use blind tests, where subjects are not aware of the aspects of the experiment
- Use big data; experimenters can use specialized algorithms in combination with multiple sets of big data:
 1. Collect detailed data into big data set
 2. Build a control group
 3. Target the best opportunities
 4. Tailor the program

Will you get maximum value from the experiment?

- Take into account the experiment's effect on customers, markets, and segments
- Ask "What works where?"
- Conduct value engineering experiments to identify product components or combination of components with the most return on investment

Product managers test product ideas to assess whether the products are likely to satisfy customers' needs. Not all ideas will deliver the performance and quality that customers expect. Therefore, product managers need to design and run experiments to see which products have the best chance of succeeding. In this course, you'll explore why product managers conduct experiments. You'll examine the best practices for conducting experiments, as well as the stages involved in conducting an experiment. Also, you'll find out how to assess whether or not an experiment is worth the time, effort, and expense. And finally, you'll explore the user-experience-based design approach.

One of the key concerns of product managers is identifying and meeting the needs of your customers. Once you have identified core needs, it's time to start generating ideas for products that meet these needs. And, as product life spans become shorter, effective product management has become a key driver for business success. How can product managers start generating product ideas? Many businesses believe that the key is to come up with ideas that are innovative. However, this doesn't always guarantee business success. Much more effective, though perhaps less exciting, are

testable ideas. Put simply, these are hypotheses that can be tested to ensure they deliver value. You need to examine the relationship between your action, the new product, and the likely outcome, the value. Because testable ideas can be scientifically analyzed to see if they work, they have built-in accountability. And this in turn encourages experimentation and practical innovation. This experimental approach to new ideas helps you understand whether something is actually a good idea in practice. You conduct experiments to test whether the idea works. Once you have assessed and validated it, you can then start to build the product. So, as well as coming up with ideas, your business needs to design experiments to test them. From a product management standpoint, experimentation is a systematic analysis of the relationship between a process and the output of that process, or between a product and the value that the product delivers. It's used to determine cause and effect. If this idea is implemented, what will be the result? It helps unearth the assumptions that underlie an idea and then validate them. However, experimentation isn't effective in every situation. You can only scientifically test an idea when the outcomes are measurable. Will this product increase sales? Will this process change reduce production time? You need to be able to measure the success of your idea. You might already be familiar with some successful product management experiments. For example, Uber wanted to test its Express Pool service. This would give users the option to walk a short distance to meet their car. The company carried out controlled experiments to test how many users would select the service and determined that the idea would be a success. Similarly, Airbnb conducted experiments on proposed new designs of its landing pages. It was able to measure search ranking and traffic and decide which designs would be most effective. By experimenting with prototypes and control groups, companies can test their ideas before fully bringing products to market. They can find the solution that delivers the greatest value, highest quality, or most reliability. Experimentation can be used throughout the product development process, changing features and functionality, testing assumptions, assessing results, and then pivoting the approach if required. And experiments can continue until the best solution is found. How can product managers use experimentation? First, it can help test existing designs, features, functionality, material, processes, and so on. Second, it can be used to develop new ideas and concepts. Remember, the ideas need to be testable with measurable outcomes. Third, using the data gathered from experimentation, product management can optimize product performance. And finally, they can improve product reliability. It's important to bear in mind that experimentation doesn't just take place during the idea generating and prototyping stages. In fact, it can be used anywhere in the product's life cycle where you can glean measurable data. For example, you can test product ideas before a launch, but then continue the experimenting after product roll out in order to improve the features. Even during the production phase, you can experiment by changing and testing various processes or product specifications. However, the more experimenting you build in early into the development process, the more likely you are to deliver the best solution. And, as you gain further insights, you can build on improvements as development progresses. There are, of course, different types of products, and there are different types of product managers. Some product managers can be regarded as builders. They lay the foundations for new solutions and conceptualize where the product needs to be. Ideation is their strong point. Maximizers, on the other hand, build on these foundations. Whereas builders often create something out of nothing, maximizers aim to maximize the value of something that has already been created. They experiment in order to test the ideas generated by the builders. Even "builder" product managers benefit from running experiments on their development projects. For example, experiments enable them to gather invaluable quantitative behavioral data about users. For example, an experiment measuring website traffic and purchases might reveal that most customers visit their website during lunch, but don't purchase until later in the evening. You can also understand who those users are. You might be assuming that customers are using your scheduling apps during work hours, but an experiment could show that they're using it more in the evening, indicating that people are using the app for more than just work-related tasks. And you can discover great insights into

how people use your product. Usability testing and control groups can enable testing assumptions about product functionality. You might think you have designed an intuitive online order form, but a usability test might reveal recurring mistakes that customers make. Challenging pre-existing assumptions is a key driver of experimenting. Experiments can yield valuable data about a product. However, product managers need to be prepared to act on this data. If an experiment reveals flaws or shortcomings in a product, you need to address them. And if an experiment doesn't yield the data needed to make an informed decision, be prepared to conduct further experiments. For example, an experiment might reveal that customers aren't happy with certain features of a new mobile device, so you need to implement changes. But you might have to conduct further experiments in order to discover exactly what those changes should be. Each experiment informs the next, in a feedback loop. If an experiment kicks off with a question, such as, why product sales drop off in the third quarter each year, the experiment should either answer that question or suggest further experiments that will answer the question. With an understanding of the experimental approach, it's then important to consider how it differs from a more traditional business case approach. In a typical business case approach, a research team brainstorms a new product idea and develops this into a business case that is presented to senior management. Once approved, the development process begins. Once the product is built, the company uses metrics to measure the success of the product. The initial idea isn't fully tested until the product is built and launched into the market. The flaws in this approach are obvious. A certain amount of guesswork comes into play, for example. The idea isn't tested before production. In other words, production starts before the viability of the idea is properly assessed. Also, it's not always clear whether senior management has the necessary knowledge to assess the idea. And it's not certain that the organization has the knowledge or resources to even deliver on the idea. As a result, problems often don't become apparent until late in the production process or, worse, when the product is released, and users start giving feedback. A huge amount of time may be spent working on a product that's ultimately abandoned due to a lack of market success. The key difference with the experimental approach is that the idea is tested before full production begins. This is why testable ideas are so important. The idea needs to yield concrete outcomes that can be rigorously measured and assessed. And experiments are used to test these ideas. Once the idea viability has been validated, senior management can approve the idea and full production can begin. Remember, however, that experimenting can continue through all stages of the production process.

Just because a product seems exciting, doesn't mean that it's going to light up the market. Product managers need to build experimenting into the product development process, and carefully analyze and act on the data that the experiments yield. This helps ensure confidence when launching the product into the marketplace. This approach obviously brings a number of benefits to the business. Rigorous testing and updating helps to improve the overall quality of the final product. Experimenting can help identify flaws early in the production process, thus reducing the product development costs. Also, a streamlined experimental schedule, involving timely fixes, can shorten product development time. The delivery of successful products at lower cost is a clear win-win scenario for any business. Product management should encourage and foster the processes, culture, resources, and capabilities needed for effective product experimentation. They must create a testing culture and ensure that senior management understands and supports the idea that ongoing testing is an integral part of the development and production process so that necessary budget and resources are available. To ensure buy-in, especially from senior management, product managers can gather evidence that shows that testing is delivering results, such as increased product success and reduced costs. The business needs to be able to conduct rigorous experiments properly and to accurately analyze the results. This involves building a testing capability, putting in place the training and resources needed to ensure effective experiments are conducted. Product managers need to understand what testing and

experimenting involve, and to convincingly advocate for them to senior management. They can even put in place an internal training program on how to design effective experiments. As you experiment, you learn. Another way to build a testing capability is to put in place processes to capture this learning, and then share the knowledge with others. Put in place a comprehensive knowledge management system in order to capture all the data you need. Retesting is an integral part of any testing process. You need to regularly revisit experiments to ensure you're capturing the information you need. When the business environment changes, you may need to revisit your tests to ensure they are still relevant. And be mindful that test data can go out of date. A final way to build a testing capability is to create a core resource group. This is comprised of experts in the field who can supervise tests and experiments. A centralized group can also ensure that groups are working in alignment and complementing each other's efforts. Having built the testing capability, the product manager then needs to ensure that they're conducting effective experiments. To do this, they need to track their information by recording data, using controls, drawing conclusions, and then reporting any errors. The aim of any experiment is to gather relevant data that will enable you to make better business decisions. You need to identify what data you need in order to test your assumption, decide how you're going to gather that data, and then present the data in a way that enables you to draw conclusions from it, using, for example, graphs or tables to show the main trends. Experiments are often used to compare one scenario against another. This is using controls. You have an experimental scenario and a control scenario. For example, if you wanted to assess the impact of a new feature on upload times, you'd compare upload times without the feature against upload times with the feature. The upload times without the feature is the control scenario. You will want to be able to draw logical conclusions from the results of your experiments. Carefully analyze the data and see what it is telling you. Often, an experiment will disprove a cherished assumption. In this case, you must still accept what the data is telling you and take the necessary actions it suggests. Finally, remember that product managers sometimes make mistakes, and you must report your errors. Maybe you included too many variables, so you can't determine which variable is affecting the results. Maybe you measured the wrong variable. Once you recognize these errors, be prepared to fix them and re-run the experiment. You can further improve the effectiveness of your experiments by testing well, using segmentation, embracing negative results, going back to the drawing board when necessary, and validating your ideas. First, how can you test well? Make sure your experiment processes are rigorous. Don't take shortcuts. Always apply rigorous methods. In other words, have a clearly defined hypothesis or question, and then gather the relevant data that addresses the hypothesis or question. Carefully analyze the data and draw logical conclusions. And, importantly, keep the control group consistent. Make sure you're analyzing the same data each time you conduct the experiment. If you change any variables, carefully note them and assess the effect that the change has on the results. Second, use segmentation. This involves digging deeper into the data to see what further learning you can unearth. This will enable you to better understand and optimize the product for different users. For example, a control group might find that 35% of users were unhappy with recent changes made to a product's user interface. However, by segmenting the data further, you might find that most of the complaints came from customers who were accessing the product using a smartphone. Next, remember that experiments don't always yield the results that you want. Learn to embrace negative results. If customers are unhappy with aspects of your product, don't blame the customer. Blame your incorrect understanding of what the customer was looking for. And after embracing the negative results, be prepared to go back to the drawing board if necessary. Get a better understanding of what the customer is looking for, and experiment with the product until it delivers what the customer really wants. Finally, validate your idea. What did the experiment tell you? How can you implement that learning? Speed is of the essence here, because if the data results show that production is moving in the wrong direction, you need to fix that as quickly as possible. But beyond these steps, there are further useful tips for conducting proper experiments. For example, you should begin with a simple proof-of-

concept experiment. This enables you to discover from the start whether your idea is even worth pursuing. Once you're satisfied that the idea is worth pursuing, you can then start refining your experiments to find out how it can be most effectively implemented. This is when you run variations of the experiment. For instance, once you've discovered that adding a specific feature improves customer experience, you can then experiment with various designs for it. Be sure to prioritize your most risky assumptions. These are assumptions that could lead to product failure if they're wrong. And if your assumptions are found to be wrong, be prepared to go back to the drawing board. Having looked at how to conduct effective experiments, it's important, then, to consider the actual experiments themselves. A typical experiment comprises several elements. First, what assumption is your experiment testing? It's better to think in terms of assumptions than ideas, because this keeps your focus on clear outcomes. If the outcomes don't align with your assumption, you'll know the assumption is wrong. Next, consider your experiment design. What experiment do you need to conduct to test your assumptions? Ensure you get any necessary stakeholder buy-in at this stage. Otherwise, you risk stakeholders questioning the results later, because they may have misunderstood what the experiment was aiming for. Also, consider the experiment participants. Who will be taking part in the experiment? Be as specific and detailed as you can. Equally important are the metrics and thresholds. What data are you measuring? What results do you need to see to know that your assumption is correct? For example, do you need to see a 5% increase in upload speed in order to justify adding that new feature to the product? Also, what metrics do you need to validate the experiment? In other words, don't over-test. When you have a clear result from the experiment, move on. And then, how will you act on the collected data? These actions need to be agreed with stakeholders. Remember, you need to act on the data findings, even if you don't get the result you like. An important concept when running experiments is that of statistical significance. Your data results need to be statistically significant in order to be valid. You need to be able to draw logical conclusions from them. And you need to carry out the experiments often enough to reasonably discount the possibility of coincidence. When you see clear patterns and trends emerging from the data, you know that the results are statistically significant, and allow you the confidence to act on them.

Developing a new product is a complex process. The product's performance depends on many variables and inter-relationships. Because of this complexity, it's important to conduct experiments to better understand the product and ensure that it is delivering what the customer needs. Conducting effective experiments involves four main stages. First, you need to ask questions and generate ideas. From these, you can create a hypothesis. Next, you conduct an experiment to test the hypothesis. And finally, you communicate your findings and close the loop. The first stage is asking questions and generating ideas. Remember, the aim of most experiments is to test a hypothesis or answer a question. In this case, the question needs to be related to the product and the customer. How can this product satisfy a customer need? Start with a goal or outcome in mind and then formulate the question around achieving that goal. For example, if the goal is to reduce customer complaints by 10%, the question could be "How can we improve the product to reduce customer complaints by 10%?" Product managers will be driving the experimentation effort, but they must make sure they involve stakeholders as well. Depending upon the experiment and the reasons for it, stakeholders may well include the team developing the product, the team tasked with marketing the product and senior management. They can ensure that your questions are aligned with broader business objectives. Involve stakeholders and make it easy for them to provide input at key stages during the experiment. The communication processes should be simple and efficient to avoid a scenario where an experiment is completed only to find that some of the key stakeholders feel the wrong question was asked. Having generated some questions, it's time to start prioritizing the ideas. What effort is involved in implementing the idea? What impact will the change have? Obviously, you're looking for maximum impact with minimum effort. Prioritize your ideas with that goal in mind. Also, bear in mind your objectives and key

results, or OKRs. Think about how the experiment aligns with the broader objectives for the product. And consider the key results that you can use to measure success. And don't forget the company's objectives and business goals. This is another part of why stakeholder involvement is essential. After asking the question, the second stage in the process is creating the hypothesis. This is where you drill-down to make the experiment more specific. A hypothesis is a testable statement that can either be proved or disproved. You must be able to collect objective data about it and evaluate that data. In general, a hypothesis examines the relationship between two items. Think of it as a cause-and-effect relationship. The experiment aims to discover the nature of that relationship. In other words, does this cause lead to that effect? A hypothesis usually has two variables and is intended to discover whether a change in one variable leads to a change in the other. One variable is independent and can be controlled. And the other variable is dependent and can be measured. Consider an example: "If we change the fuel type, then the engine efficiency will increase." In this case, the fuel type is the independent variable, because the fuel efficiency depends on the fuel type. In the experiment, you can control the fuel type, testing the results from different fuel types. The fuel efficiency on the other hand, is the dependent variable. When you change the independent variable, you change the dependent variable. In this case when you change the fuel type, you change the fuel efficiency, and in the experiment you measure that change. A hypothesis usually takes the form of an if, then statement. The if clause relates to the controlled independent variable, and the then clause relates to the measured dependent variable. As you control the independent variable by changing it, you can measure the impact on the dependent variable. So, in this example, as you control or change the fuel type, you can measure the impact on the fuel efficiency, and then you can check whether your hypothesis is correct. As you begin your experiment, you are likely to generate a number of ideas and then hypotheses. So how do you decide which ones to test? First, make sure that you can objectively test the hypothesis. Can you get measurable data about it? Is it easy to find compelling evidence? Also, think about what evidence already exists. Perhaps similar experiments have been conducted before. Then consider the impact on the product if the hypothesis is true. And see how this impact will affect your overall business goals. Once you've generated ideas and formulated a hypothesis, it's time to conduct the experiment. There are many different types of experiment available, but the two most common are A/B testing and multivariate testing. A/B testing involves comparing one or more variants against the control. Consider the fuel example, again. You might set up two experiments, one with the existing fuel type and one with a new fuel type. The existing fuel type is the control group, or group A, and the new fuel type is the experimental group, or group B. You would measure the fuel efficiency in both groups and compare the results. So, you're comparing the results of groups A and B. Multivariate testing is similar, but as the name suggests, it involves multiple variants. So instead of having just examined how the fuel type affects fuel efficiency, you might also look at how engine size and body weight affect the fuel efficiency. Obviously, multivariate testing is much more complex to conduct since you need to control for multiple variables. It would be easy to get caught up in an endless cycle of experiments, testing countless variables. But product managers need to ensure they're not wasting money, time, or resources on experiments, that won't deliver the data they need to make business decisions. They must decide which experiments to conduct. Sometimes, good enough is good enough. You don't always have to aim for perfection. Perhaps a minimum viable product is all that's needed. What's needed is a product you can experiment on that provides the information needed to create a product that satisfies customers. Some experiments require more effort than others. If the product team requires results fast, it makes sense to prioritize experiments that involve minimum work and deliver quick data. Remember that you don't always need a final product to experiment on. A functional prototype might be all that's needed to test some key hypotheses. The prototype might also alert you to any potential problems with the finished product. After conducting the experiment, it's time to communicate its findings and then close the loops. Did the experiment deliver enough data to prove or disprove the hypothesis? Did it throw up any unexpected

results? You need to carefully examine all the data and update the stakeholders, so that everyone knows what story the experiment is telling. When communicating, present the findings in easy-to-understand, digestible formats, such as tables or graphs. Don't make stakeholders or team members wade through a sea of numbers. Highlight the key trends and present the evidence as concisely as possible. Stakeholders will want to know what the findings mean for future product developments. Projects may have to be prioritized or scrapped, depending on the findings. They may also want to know why the experiments delivered these findings. Perhaps further investigation will be needed to uncover root causes. Also, be sure to bring the process to closure. Outline the next steps, identify the lessons learned, and lay out the actions required. If the experiment shows that something doesn't work, outline steps to fix the problem. And if it shows that something does work, outline steps to implement this change. This is particularly important for stakeholders, so that they can see tangible results from the experiments. And be sure to record and store all your findings.

Conducting business experiments can have a significant payoff. However, they also require a significant investment in time, expense, and effort. Before launching an experiment, product managers need to ask some key questions to ensure the experiment will be worth the investment. First, does the experiment have a clear purpose? When trying to answer business questions, you need to determine whether experiments are the best practical way to address the problem. To do this, first consider what you need to learn in order to answer the question. You must have a clear hypothesis. Obviously, in some cases, running an experiment will indeed be the best approach to addressing the business problem. Once you decide this, you should start thinking about the scope of the experiment. You need to conduct enough experiments to address the problem, but no more experiments than required. Also, consider the ancillary effects of an experiment. Making one change to a product feature set may have unintended consequences, either negative or positive. The second question is: Have stakeholders made a commitment to abide by the results? Stakeholder agreement is crucial before starting any experiment. You need to agree with stakeholders on how the organization will proceed once the results of the experiment are clear. Stakeholders must commit to taking all results into account. They cannot simply pick the data that they feel confirms their own pre-existing biases. Sometimes, no matter how enthusiastic everyone is about an exciting initiative, you have to cancel a project. The experiment results might show that the initiative is not worth pursuing. Stakeholders must be willing to walk away from a project, no matter how committed they originally were to it. Indeed, it's a good idea to put a process in place to ensure all results are considered. And make sure you have senior executive buy-in when necessary. Of course, to encourage this buy-in, your experiments need to be aligned to the organization's overall business objectives. This helps to ensure stakeholder engagement, because they can see how the experiment contributes to the success of the organization. The third question is: Is the experiment doable? The aim of experiments is to find out whether a project is viable. By testing different variables you can determine what impact, if any, various changes could have on the success of the project. However, in any project, there are likely to be a large number of variables, and a large number of interactions between them. This makes it difficult to assess whether there's a firm, cause-and-effect relationship between any given variable and a result. To address this concern, it's very important to get the experiment sample size right. With the right sample, your experiment results are more likely to be statistically valid. Also, by optimizing the sample size, you can reduce the cost of the experiment, because your samples will be no larger than they need to be. And finally, the more reliable your experiments become, because of accurate sample sizes, the more effectively your organization can innovate. However, when considering sample size, and the trade-off between size and cost, you need to keep a few things in mind. Using larger samples can help you better understand the relationships between different variables. Despite this benefit, there are cost implications associated with running experiments using larger sample sizes, and so it's not always a viable option. Also, large samples do not

necessarily guarantee better data. Also, bear in mind that when causes have a smaller impact, you need to use larger sample sizes. This is because it can be more difficult to determine the cause of a small change. If the impact is likely to be large, however, you can use a smaller sample. This leads us on to the next question: Can you ensure reliable results from the experiments you run? Obviously, it's important that your experiments yield reliable results. However, product managers have to consider other issues, such as cost, time, and other practical considerations, such as the availability of people and equipment to even conduct the experiment. As a product manager, you can use common methods to improve the reliability of your results. These are randomized field trials, blind tests, and data analysis. The first method, randomized field trials, involve using two similar sample groups. A change is applied to one group, the test group. And a control, or untested group, does not have the change. This enables you to compare the results between the test group and the control group. If the difference in results is significantly different, you can link that difference to the change in the test group. By using randomized testing, you can reduce systemic bias. In other words, you reduce the risk of people selecting samples that will give the results that they are hoping for. This in turn makes it easier to determine the cause of the different results between the two samples. The second method is blind tests. Experiments are susceptible to what's known as the Hawthorne Effect. This occurs when experiment participants alter the behavior, often without realizing it, because they know they are part of an experiment. In a blind test, participants are not aware that an experiment is taking place. For example, a company might alter a component of a product without users being aware of the change. The third method to increase the reliability of experiment results is to use large volumes of data, and apply complex algorithms to analyze these multiple datasets. The results of this analysis will enable you to identify which changes have the most significant impact on the product's success. Your final question is: Will you get maximum value from the experiment? The costs involved in running experiments can be considerable, so you want to ensure that you get a good return on your investment of time and money. In order to get a good return on investment, you need to understand if the experiment will provide insight into the product's impact on customers, markets, and customer segments to suggest where the product is likely to find its greatest chance of success. By carefully analyzing and then exploiting the insights gained from your experiments, you can ensure that the experiments deliver value to the organization and the customers. Indeed, you should spend at least as much, if not more, time analyzing the data, and gathering key insights as you do collecting the data.

As users think about or interact with a product, they experience responses, both positive and negative. These responses are referred to as the user experience, or UX. The International Organization for Standardization defines UX as, "persons' perceptions and responses resulting from the use and/or anticipated use of a product, system or service". By taking into account UX, product managers can design product experiments aimed at increasing users' positive experiences with the product, and decreasing their negative ones. By systematically capturing user responses, you can feed this information into future design decisions to drive better product delivery. If a current product isn't meeting user's expectations, you need to find out why and rectify the situation. Ideally, user responses should be gathered as soon as possible in the planning and production phases, so that they can be considered as the product is developed. A key concept in user experience-based design of experiments, or UXDoE philosophy is that customer satisfaction plays a critical role in product quality. In other words, a quality product is one that satisfies customers and meets their expectations. There is sometimes a disconnect between what product designers think customers want, and what customers actually want. UXDoE enables you to carry out user research to bridge this gap. You try to gather as much information about the customers as possible, including psychological, lifestyle, demographic, and technology factors. This information in turn drives product development effort. This is a very efficient approach to product development because it enables you to quickly define users' needs and expectations, and then build the

product that will meet these needs. All product development takes place in the context of understanding what the user expects, thus reducing the chances of building a product that doesn't meet those expectations. A typical UXDoE process involves three steps. First, you define the problem. Next, you identify user response and experimental factor and finally, you execute the experiment. Defining the problem involves establishing a link between the user's experience and the product design. Of course, you could collect data from almost limitless sources, with users having diverse backgrounds and experience. So, at this stage, you need to define which users you should focus on and then define the problem in terms of meeting their expectations. You also need a means of identifying and measuring user responses and experimental factors. The experimental factor is the variable that you're focusing on, such as upload speed, visual appeal, or ease of interface navigation, for example. You might decide on using a rating system that users complete in order to record their satisfaction with different aesthetic elements of the product. Having defined the problem and how you're going to measure it, you can execute the experiment. This involves determining what form the experiment is going to take and who will participate in it. Product managers need to decide issues such as the scope of the experiment, the sequence of the tests, the management of the experiment environment, the collection of data, and the measurement of results. They may need to conduct some trial rounds before deciding on a final process. And once the experiment is up and running, it should be carefully monitored. In order to understand user responses, product managers need to understand the users. They need to find out about their psychology and life factors. Are they comfortable using new products or do they resist change? Are they risk averse? Depending on the product, they may need to understand customer's attitude toward and acceptance of new technologies. Are they able to adapt to new technologies easily, or do they need a certain amount of support and coaching? Also, bear in mind their lifestyle and persona. Is the user a single mother on a tight budget who is looking for a bargain, or a retiree who likes to indulge in luxury products? And don't forget demographic variables. These include age, location, occupation, income, and so on. So how does UXDoE impact product design and development? First, it helps to clarify the conceptual activities that need to be carried out before building the product. The customers' needs can drive the concepts that inform how the product is built. Once you have clarified the underlying concepts, you can begin teasing out the engineering details needed to turn the concepts into a concrete product that delivers what customers want. Using UXDoE, product designers can identify subjects who can take part in the experiments. These are the users who can provide the most valuable feedback about the product. UXDoE also helps designers decide which experimental factors to investigate. These are the product variables that can be measured to see their impacts on product performance and quality. And UXDoE enables the designers to then allocate the resources needed for the experiments. Designers and product managers can work together to decide budgets, team members, equipment, and other resources required. Next, you need to work with stakeholders to put in place the processes required to implement these engineering details. This involves building and checking the product, ensuring there are processes that enhance the quality of the finished product. In most cases, you'll start by manufacturing a prototype. This enables you to get further feedback before diving into full-scale production. Of course, user feedback doesn't end with the launch of the product. Product managers must put in place post-launch activities that enable them to gather further information about the product. They can use this post-launch feedback to refine the product and address any quality issues that come to light. UXDoE benefits product managers by providing vital, early feedback about the product, its design, and usability. It allows product managers to evaluate the products while they are still in the conceptual development and testing stages. Product managers also gain a deeper understanding of the products themselves. By studying the user-product interaction, you can see which features are most popular with users. This helps determine what the key features are, which in turn can then drive not only the product design, but future marketing efforts. By getting a better understanding of which type of users are most positive about the product, product managers can

identify target groups and develop marketing strategies around those audiences to maximize the product success in the marketplace.

1. Video: Product Management: Designing and Running Experiments (bs_apm01_a07_enus_01)



After completing this video, you will be able to discover the subject areas that will be covered in this course.

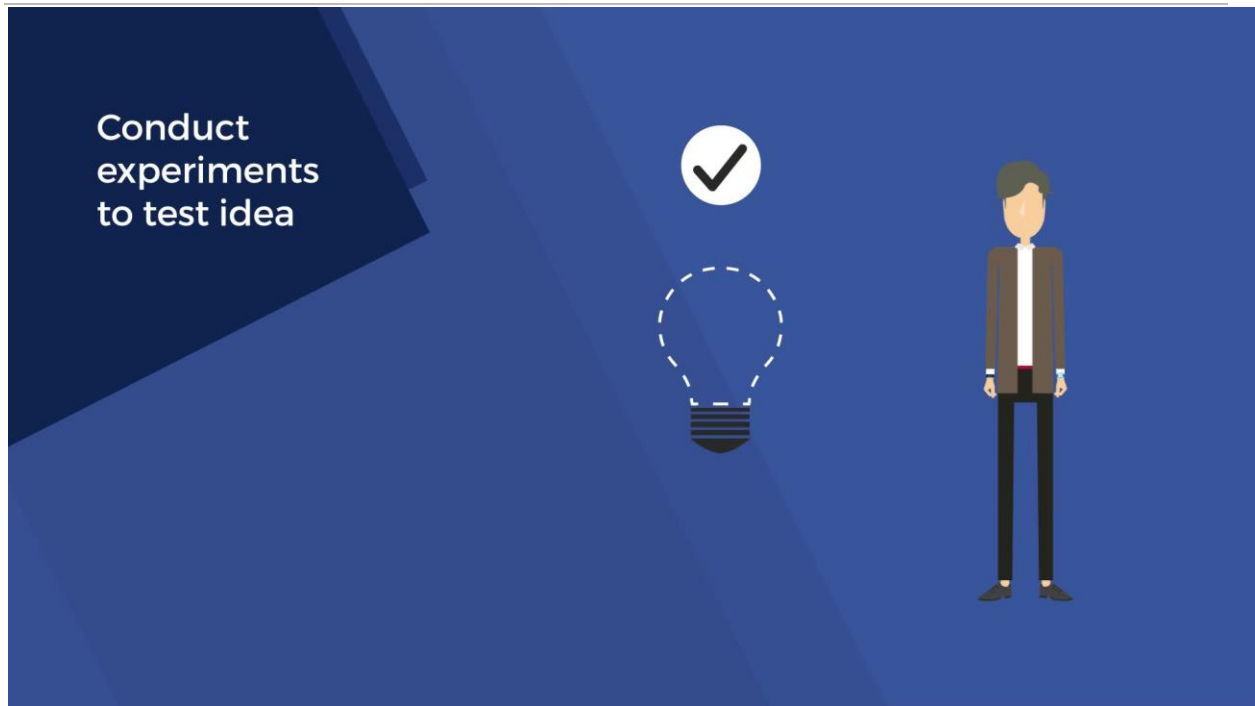
Objectives

- *discover the subject areas that will be covered in this course*

[Video description begins] *Course title: Product Management: Designing and Running Experiments.* [Video description ends]

Product managers test product ideas to assess whether the products are likely to satisfy customers' needs. Not all ideas will deliver the performance and quality that customers expect. Therefore, product managers need to design and run experiments to see which products have the best chance of succeeding. In this course, you'll explore why product managers conduct experiments. You'll examine the best practices for conducting experiments, as well as the stages involved in conducting an experiment. Also, you'll find out how to assess whether or not an experiment is worth the time, effort, and expense. And finally, you'll explore the user-experience-based design approach.

2. Video: Make Better Product Decisions with Experimentation (bs_apm01_a07_enus_02)



After completing this video, you will be able to identify the reasons why product managers run experiments

Objectives

- *identify the reasons why product managers run experiments*

[Video description begins] *Topic title: Make Better Product Decisions with Experimentation.* [Video description ends]

One of the key concerns of product managers is identifying and meeting the needs of your customers. Once you have identified core needs, it's time to start generating ideas for products that meet these needs. And, as product life spans become shorter, effective product management has become a key driver for business success. How can product managers start generating product ideas? Many businesses believe that the key is to come up with ideas that are innovative. However, this doesn't always guarantee business success. Much more effective, though perhaps less exciting, are testable ideas. Put simply, these are hypotheses that can be tested to ensure they deliver value.

You need to examine the relationship between your action, the new product, and the likely outcome, the value. Because testable ideas can be scientifically analyzed to see if they work, they have built-in accountability. And this in turn encourages experimentation and practical innovation. This experimental approach to new ideas helps you understand whether something is actually a good idea in practice. You conduct experiments to test whether the idea works. Once you have assessed and

validated it, you can then start to build the product. So, as well as coming up with ideas, your business needs to design experiments to test them.

From a product management standpoint, experimentation is a systematic analysis of the relationship between a process and the output of that process, or between a product and the value that the product delivers. It's used to determine cause and effect. If this idea is implemented, what will be the result? It helps unearth the assumptions that underlie an idea and then validate them. However, experimentation isn't effective in every situation. You can only scientifically test an idea when the outcomes are measurable. Will this product increase sales? Will this process change reduce production time? You need to be able to measure the success of your idea. You might already be familiar with some successful product management experiments.

For example, Uber wanted to test its Express Pool service. This would give users the option to walk a short distance to meet their car. The company carried out controlled experiments to test how many users would select the service and determined that the idea would be a success. Similarly, Airbnb conducted experiments on proposed new designs of its landing pages. It was able to measure search ranking and traffic and decide which designs would be most effective. By experimenting with prototypes and control groups, companies can test their ideas before fully bringing products to market. They can find the solution that delivers the greatest value, highest quality, or most reliability.

Experimentation can be used throughout the product development process, changing features and functionality, testing assumptions, assessing results, and then pivoting the approach if required. And experiments can continue until the best solution is found. How can product managers use experimentation? First, it can help test existing designs, features, functionality, material, processes, and so on. Second, it can be used to develop new ideas and concepts. Remember, the ideas need to be testable with measurable outcomes. Third, using the data gathered from experimentation, product management can optimize product performance. And finally, they can improve product reliability. It's important to bear in mind that experimentation doesn't just take place during the idea generating and prototyping stages.

In fact, it can be used anywhere in the product's life cycle where you can glean measurable data. For example, you can test product ideas before a launch, but then continue the experimenting after product roll out in order to improve the features. Even during the production phase, you can experiment by changing and testing various processes or product specifications. However, the more experimenting you build in early into the development process, the more likely you are to deliver the best solution. And, as you gain further insights, you can build on improvements as

development progresses. There are, of course, different types of products, and there are different types of product managers.

Some product managers can be regarded as builders. They lay the foundations for new solutions and conceptualize where the product needs to be. Ideation is their strong point. Maximizers, on the other hand, build on these foundations. Whereas builders often create something out of nothing, maximizers aim to maximize the value of something that has already been created. They experiment in order to test the ideas generated by the builders. Even "builder" product managers benefit from running experiments on their development projects. For example, experiments enable them to gather invaluable quantitative behavioral data about users. For example, an experiment measuring website traffic and purchases might reveal that most customers visit their website during lunch, but don't purchase until later in the evening. You can also understand who those users are. You might be assuming that customers are using your scheduling apps during work hours, but an experiment could show that they're using it more in the evening, indicating that people are using the app for more than just work-related tasks. And you can discover great insights into how people use your product. Usability testing and control groups can enable testing assumptions about product functionality.

You might think you have designed an intuitive online order form, but a usability test might reveal recurring mistakes that customers make. Challenging pre-existing assumptions is a key driver of experimenting. Experiments can yield valuable data about a product. However, product managers need to be prepared to act on this data. If an experiment reveals flaws or shortcomings in a product, you need to address them. And if an experiment doesn't yield the data needed to make an informed decision, be prepared to conduct further experiments. For example, an experiment might reveal that customers aren't happy with certain features of a new mobile device, so you need to implement changes. But you might have to conduct further experiments in order to discover exactly what those changes should be.

Each experiment informs the next, in a feedback loop. If an experiment kicks off with a question, such as, why product sales drop off in the third quarter each year, the experiment should either answer that question or suggest further experiments that will answer the question. With an understanding of the experimental approach, it's then important to consider how it differs from a more traditional business case approach. In a typical business case approach, a research team brainstorms a new product idea and develops this into a business case that is presented to senior management. Once approved, the development process begins.

Once the product is built, the company uses metrics to measure the success of the product. The initial idea isn't fully tested until the product is built and launched into the market. The flaws in this approach are obvious. A certain amount of guesswork comes into play, for example. The idea isn't tested before production. In other words,

production starts before the viability of the idea is properly assessed. Also, it's not always clear whether senior management has the necessary knowledge to assess the idea. And it's not certain that the organization has the knowledge or resources to even deliver on the idea.

As a result, problems often don't become apparent until late in the production process or, worse, when the product is released, and users start giving feedback. A huge amount of time may be spent working on a product that's ultimately abandoned due to a lack of market success. The key difference with the experimental approach is that the idea is tested before full production begins. This is why testable ideas are so important. The idea needs to yield concrete outcomes that can be rigorously measured and assessed. And experiments are used to test these ideas. Once the idea viability has been validated, senior management can approve the idea and full production can begin. Remember, however, that experimenting can continue through all stages of the production process.

3. Video: Best Practices for Effective Product Experiments (bs_apm01_a07_enus_03)



After completing this video, you will be able to identify the steps for ensuring an effective experiment

Objectives

- *identify the steps for ensuring an effective experiment*

[Video description begins] *Topic title: Best Practices for Effective Product Experiments.* [Video description ends]

Just because a product seems exciting, doesn't mean that it's going to light up the market. Product managers need to build experimenting into the product development process, and carefully analyze and act on the data that the experiments yield. This helps ensure confidence when launching the product into the marketplace. This approach obviously brings a number of benefits to the business. Rigorous testing and updating helps to improve the overall quality of the final product. Experimenting can help identify flaws early in the production process, thus reducing the product development costs. Also, a streamlined experimental schedule, involving timely fixes, can shorten product development time.

The delivery of successful products at lower cost is a clear win-win scenario for any business. Product management should encourage and foster the processes, culture, resources, and capabilities needed for effective product experimentation. They must create a testing culture and ensure that senior management understands and supports the idea that ongoing testing is an integral part of the development and production process so that necessary budget and resources are available. To ensure buy-in, especially from senior management, product managers can gather evidence that shows that testing is delivering results, such as increased product success and reduced costs.

The business needs to be able to conduct rigorous experiments properly and to accurately analyze the results. This involves building a testing capability, putting in place the training and resources needed to ensure effective experiments are conducted. Product managers need to understand what testing and experimenting involve, and to convincingly advocate for them to senior management. They can even put in place an internal training program on how to design effective experiments. As you experiment, you learn. Another way to build a testing capability is to put in place processes to capture this learning, and then share the knowledge with others. Put in place a comprehensive knowledge management system in order to capture all the data you need.

Retesting is an integral part of any testing process. You need to regularly revisit experiments to ensure you're capturing the information you need. When the business environment changes, you may need to revisit your tests to ensure they are still relevant. And be mindful that test data can go out of date. A final way to build a testing capability is to create a core resource group. This is comprised of experts in the field who can supervise tests and experiments. A centralized group can also ensure that groups are working in alignment and complementing each other's efforts.

Having built the testing capability, the product manager then needs to ensure that they're conducting effective experiments. To do this, they need to track their information by recording data, using controls, drawing conclusions, and then reporting any errors. The aim of any experiment is to gather relevant data that will

enable you to make better business decisions. You need to identify what data you need in order to test your assumption, decide how you're going to gather that data, and then present the data in a way that enables you to draw conclusions from it, using, for example, graphs or tables to show the main trends.

Experiments are often used to compare one scenario against another. This is using controls. You have an experimental scenario and a control scenario. For example, if you wanted to assess the impact of a new feature on upload times, you'd compare upload times without the feature against upload times with the feature. The upload times without the feature is the control scenario. You will want to be able to draw logical conclusions from the results of your experiments. Carefully analyze the data and see what it is telling you. Often, an experiment will disprove a cherished assumption. In this case, you must still accept what the data is telling you and take the necessary actions it suggests.

Finally, remember that product managers sometimes make mistakes, and you must report your errors. Maybe you included too many variables, so you can't determine which variable is affecting the results. Maybe you measured the wrong variable. Once you recognize these errors, be prepared to fix them and re-run the experiment. You can further improve the effectiveness of your experiments by testing well, using segmentation, embracing negative results, going back to the drawing board when necessary, and validating your ideas. First, how can you test well? Make sure your experiment processes are rigorous.

Don't take shortcuts. Always apply rigorous methods. In other words, have a clearly defined hypothesis or question, and then gather the relevant data that addresses the hypothesis or question. Carefully analyze the data and draw logical conclusions. And, importantly, keep the control group consistent. Make sure you're analyzing the same data each time you conduct the experiment. If you change any variables, carefully note them and assess the effect that the change has on the results. Second, use segmentation. This involves digging deeper into the data to see what further learning you can unearth. This will enable you to better understand and optimize the product for different users.

For example, a control group might find that 35% of users were unhappy with recent changes made to a product's user interface. However, by segmenting the data further, you might find that most of the complaints came from customers who were accessing the product using a smartphone. Next, remember that experiments don't always yield the results that you want. Learn to embrace negative results. If customers are unhappy with aspects of your product, don't blame the customer. Blame your incorrect understanding of what the customer was looking for. And after embracing the negative results, be prepared to go back to the drawing board if necessary. Get a better understanding of what the customer is looking for, and experiment with the product until it delivers what the customer really wants.

Finally, validate your idea. What did the experiment tell you? How can you implement that learning? Speed is of the essence here, because if the data results show that production is moving in the wrong direction, you need to fix that as quickly as possible. But beyond these steps, there are further useful tips for conducting proper experiments. For example, you should begin with a simple proof-of-concept experiment. This enables you to discover from the start whether your idea is even worth pursuing.

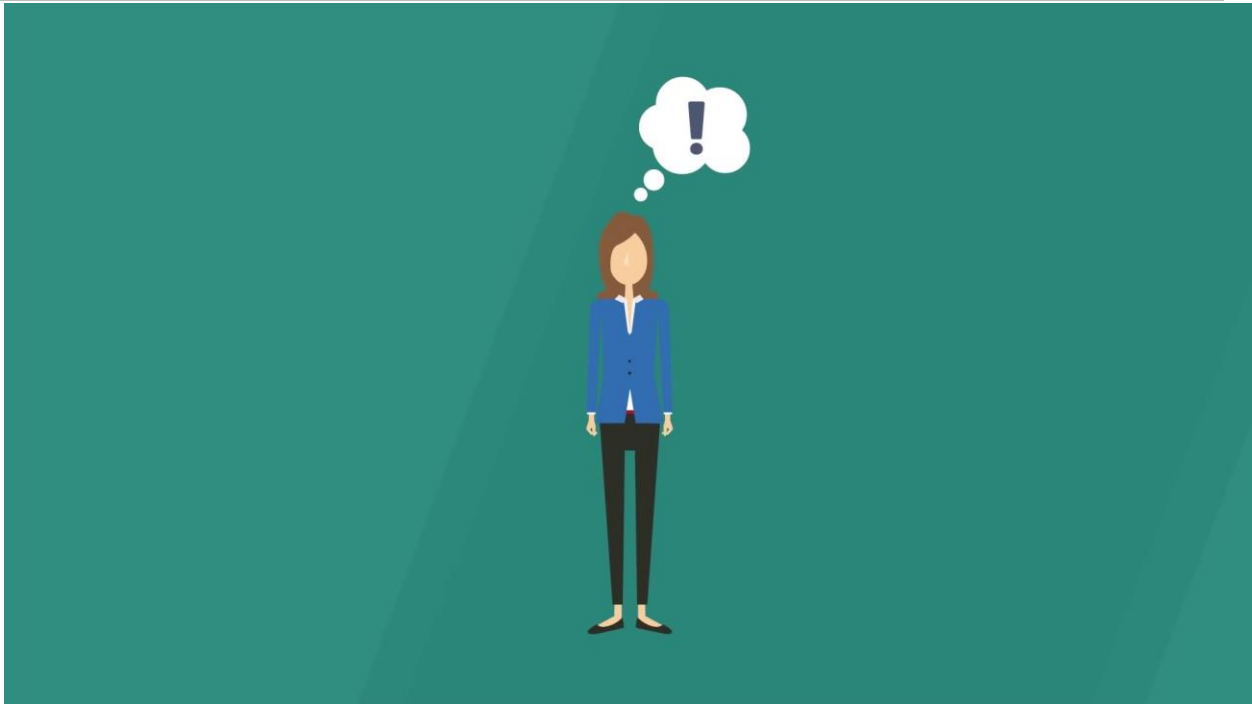
Once you're satisfied that the idea is worth pursuing, you can then start refining your experiments to find out how it can be most effectively implemented. This is when you run variations of the experiment. For instance, once you've discovered that adding a specific feature improves customer experience, you can then experiment with various designs for it. Be sure to prioritize your most risky assumptions. These are assumptions that could lead to product failure if they're wrong. And if your assumptions are found to be wrong, be prepared to go back to the drawing board. Having looked at how to conduct effective experiments, it's important, then, to consider the actual experiments themselves. A typical experiment comprises several elements.

First, what assumption is your experiment testing? It's better to think in terms of assumptions than ideas, because this keeps your focus on clear outcomes. If the outcomes don't align with your assumption, you'll know the assumption is wrong. Next, consider your experiment design. What experiment do you need to conduct to test your assumptions? Ensure you get any necessary stakeholder buy-in at this stage. Otherwise, you risk stakeholders questioning the results later, because they may have misunderstood what the experiment was aiming for.

Also, consider the experiment participants. Who will be taking part in the experiment? Be as specific and detailed as you can. Equally important are the metrics and thresholds. What data are you measuring? What results do you need to see to know that your assumption is correct? For example, do you need to see a 5% increase in upload speed in order to justify adding that new feature to the product? Also, what metrics do you need to validate the experiment? In other words, don't over-test. When you have a clear result from the experiment, move on. And then, how will you act on the collected data? These actions need to be agreed with stakeholders. Remember, you need to act on the data findings, even if you don't get the result you like. An important concept when running experiments is that of statistical significance.

Your data results need to be statistically significant in order to be valid. You need to be able to draw logical conclusions from them. And you need to carry out the experiments often enough to reasonably discount the possibility of coincidence. When you see clear patterns and trends emerging from the data, you know that the results are statistically significant, and allow you the confidence to act on them.

4. Video: The Stages of Conducting a Product Experiment (bs_apm01_a07_enus_04)



After completing this video, you will be able to sequence the stages for conducting an experiment

Objectives

- *sequence the stages for conducting an experiment*

[Video description begins] *Topic title: The Stages of Conducting a Product Experiment.* [Video description ends]

Developing a new product is a complex process. The product's performance depends on many variables and inter-relationships. Because of this complexity, it's important to conduct experiments to better understand the product and ensure that it is delivering what the customer needs. Conducting effective experiments involves four main stages. First, you need to ask questions and generate ideas. From these, you can create a hypothesis. Next, you conduct an experiment to test the hypothesis. And finally, you communicate your findings and close the loop. The first stage is asking questions and generating ideas.

Remember, the aim of most experiments is to test a hypothesis or answer a question. In this case, the question needs to be related to the product and the customer. How can this product satisfy a customer need? Start with a goal or outcome in mind and then formulate the question around achieving that goal. For example, if the goal is to reduce customer complaints by 10%, the question could be "How can we improve the product to reduce customer complaints by 10%?" Product

managers will be driving the experimentation effort, but they must make sure they involve stakeholders as well.

Depending upon the experiment and the reasons for it, stakeholders may well include the team developing the product, the team tasked with marketing the product and senior management. They can ensure that your questions are aligned with broader business objectives. Involve stakeholders and make it easy for them to provide input at key stages during the experiment. The communication processes should be simple and efficient to avoid a scenario where an experiment is completed only to find that some of the key stakeholders feel the wrong question was asked. Having generated some questions, it's time to start prioritizing the ideas. What effort is involved in implementing the idea? What impact will the change have? Obviously, you're looking for maximum impact with minimum effort. Prioritize your ideas with that goal in mind.

Also, bear in mind your objectives and key results, or OKRs. Think about how the experiment aligns with the broader objectives for the product. And consider the key results that you can use to measure success. And don't forget the company's objectives and business goals. This is another part of why stakeholder involvement is essential. After asking the question, the second stage in the process is creating the hypothesis. This is where you drill-down to make the experiment more specific. A hypothesis is a testable statement that can either be proved or disproved. You must be able to collect objective data about it and evaluate that data. In general, a hypothesis examines the relationship between two items. Think of it as a cause-and-effect relationship. The experiment aims to discover the nature of that relationship.

In other words, does this cause lead to that effect? A hypothesis usually has two variables and is intended to discover whether a change in one variable leads to a change in the other. One variable is independent and can be controlled. And the other variable is dependent and can be measured. Consider an example: "If we change the fuel type, then the engine efficiency will increase." In this case, the fuel type is the independent variable, because the fuel efficiency depends on the fuel type. In the experiment, you can control the fuel type, testing the results from different fuel types. The fuel efficiency on the other hand, is the dependent variable. When you change the independent variable, you change the dependent variable. In this case when you change the fuel type, you change the fuel efficiency, and in the experiment you measure that change. A hypothesis usually takes the form of an if, then statement. The if clause relates to the controlled independent variable, and the then clause relates to the measured dependent variable. As you control the independent variable by changing it, you can measure the impact on the dependent variable. So, in this example, as you control or change the fuel type, you

can measure the impact on the fuel efficiency, and then you can check whether your hypothesis is correct.

As you begin your experiment, you are likely to generate a number of ideas and then hypotheses. So how do you decide which ones to test? First, make sure that you can objectively test the hypothesis. Can you get measurable data about it? Is it easy to find compelling evidence? Also, think about what evidence already exists. Perhaps similar experiments have been conducted before. Then consider the impact on the product if the hypothesis is true. And see how this impact will affect your overall business goals. Once you've generated ideas and formulated a hypothesis, it's time to conduct the experiment.

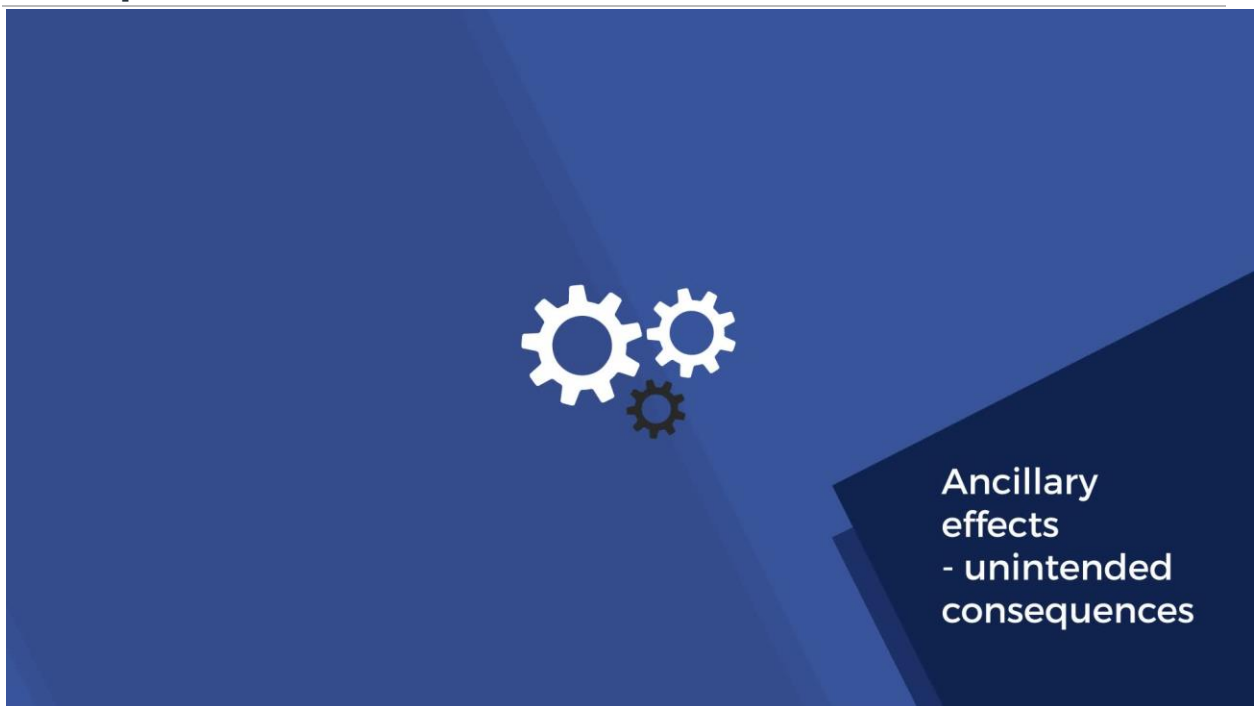
There are many different types of experiment available, but the two most common are A/B testing and multivariate testing. A/B testing involves comparing one or more variants against the control. Consider the fuel example, again. You might set up two experiments, one with the existing fuel type and one with a new fuel type. The existing fuel type is the control group, or group A, and the new fuel type is the experimental group, or group B. You would measure the fuel efficiency in both groups and compare the results. So, you're comparing the results of groups A and B. Multivariate testing is similar, but as the name suggests, it involves multiple variants. So instead of having just examined how the fuel type affects fuel efficiency, you might also look at how engine size and body weight affect the fuel efficiency.

Obviously, multivariate testing is much more complex to conduct since you need to control for multiple variables. It would be easy to get caught up in an endless cycle of experiments, testing countless variables. But product managers need to ensure they're not wasting money, time, or resources on experiments, that won't deliver the data they need to make business decisions. They must decide which experiments to conduct. Sometimes, good enough is good enough. You don't always have to aim for perfection. Perhaps a minimum viable product is all that's needed. What's needed is a product you can experiment on that provides the information needed to create a product that satisfies customers.

Some experiments require more effort than others. If the product team requires results fast, it makes sense to prioritize experiments that involve minimum work and deliver quick data. Remember that you don't always need a final product to experiment on. A functional prototype might be all that's needed to test some key hypotheses. The prototype might also alert you to any potential problems with the finished product. After conducting the experiment, it's time to communicate its findings and then close the loops. Did the experiment deliver enough data to prove or disprove the hypothesis? Did it throw up any unexpected results? You need to carefully examine all the data and update the stakeholders, so that everyone knows what story the experiment is telling.

When communicating, present the findings in easy-to-understand, digestible formats, such as tables or graphs. Don't make stakeholders or team members wade through a sea of numbers. Highlight the key trends and present the evidence as concisely as possible. Stakeholders will want to know what the findings mean for future product developments. Projects may have to be prioritized or scrapped, depending on the findings. They may also want to know why the experiments delivered these findings. Perhaps further investigation will be needed to uncover root causes. Also, be sure to bring the process to closure. Outline the next steps, identify the lessons learned, and lay out the actions required. If the experiment shows that something doesn't work, outline steps to fix the problem. And if it shows that something does work, outline steps to implement this change. This is particularly important for stakeholders, so that they can see tangible results from the experiments. And be sure to record and store all your findings.

5. Video: Running Experiments That Are Worth the Investment (bs_apm01_a07_enus_05)



After completing this video, you will be able to identify the questions to ask to see if an experiment is worth the expense and effort

Objectives

- *identify the questions to ask to see if an experiment is worth the expense and effort*

[Video description begins] *Topic title: Running Experiments That Are Worth the Investment.* [Video description ends]

Conducting business experiments can have a significant payoff. However, they also require a significant investment in time, expense, and effort. Before launching an experiment, product managers need to ask some key questions to ensure the experiment will be worth the investment. First, does the experiment have a clear purpose? When trying to answer business questions, you need to determine whether experiments are the best practical way to address the problem. To do this, first consider what you need to learn in order to answer the question. You must have a clear hypothesis.

Obviously, in some cases, running an experiment will indeed be the best approach to addressing the business problem. Once you decide this, you should start thinking about the scope of the experiment. You need to conduct enough experiments to address the problem, but no more experiments than required. Also, consider the ancillary effects of an experiment. Making one change to a product feature set may have unintended consequences, either negative or positive. The second question is: Have stakeholders made a commitment to abide by the results? Stakeholder agreement is crucial before starting any experiment. You need to agree with stakeholders on how the organization will proceed once the results of the experiment are clear.

Stakeholders must commit to taking all results into account. They cannot simply pick the data that they feel confirms their own pre-existing biases. Sometimes, no matter how enthusiastic everyone is about an exciting initiative, you have to cancel a project. The experiment results might show that the initiative is not worth pursuing. Stakeholders must be willing to walk away from a project, no matter how committed they originally were to it. Indeed, it's a good idea to put a process in place to ensure all results are considered. And make sure you have senior executive buy-in when necessary.

Of course, to encourage this buy-in, your experiments need to be aligned to the organization's overall business objectives. This helps to ensure stakeholder engagement, because they can see how the experiment contributes to the success of the organization. The third question is: Is the experiment doable? The aim of experiments is to find out whether a project is viable. By testing different variables you can determine what impact, if any, various changes could have on the success of the project.

However, in any project, there are likely to be a large number of variables, and a large number of interactions between them. This makes it difficult to assess whether there's a firm, cause-and-effect relationship between any given variable and a result. To address this concern, it's very important to get the experiment sample size right. With the right sample, your experiment results are more likely to be statistically valid. Also, by optimizing the sample size, you can reduce the cost of the experiment, because your samples will be no larger than they need to be. And finally, the more

reliable your experiments become, because of accurate sample sizes, the more effectively your organization can innovate.

However, when considering sample size, and the trade-off between size and cost, you need to keep a few things in mind. Using larger samples can help you better understand the relationships between different variables. Despite this benefit, there are cost implications associated with running experiments using larger sample sizes, and so it's not always a viable option. Also, large samples do not necessarily guarantee better data. Also, bear in mind that when causes have a smaller impact, you need to use larger sample sizes. This is because it can be more difficult to determine the cause of a small change. If the impact is likely to be large, however, you can use a smaller sample. This leads us on to the next question: Can you ensure reliable results from the experiments you run? Obviously, it's important that your experiments yield reliable results.

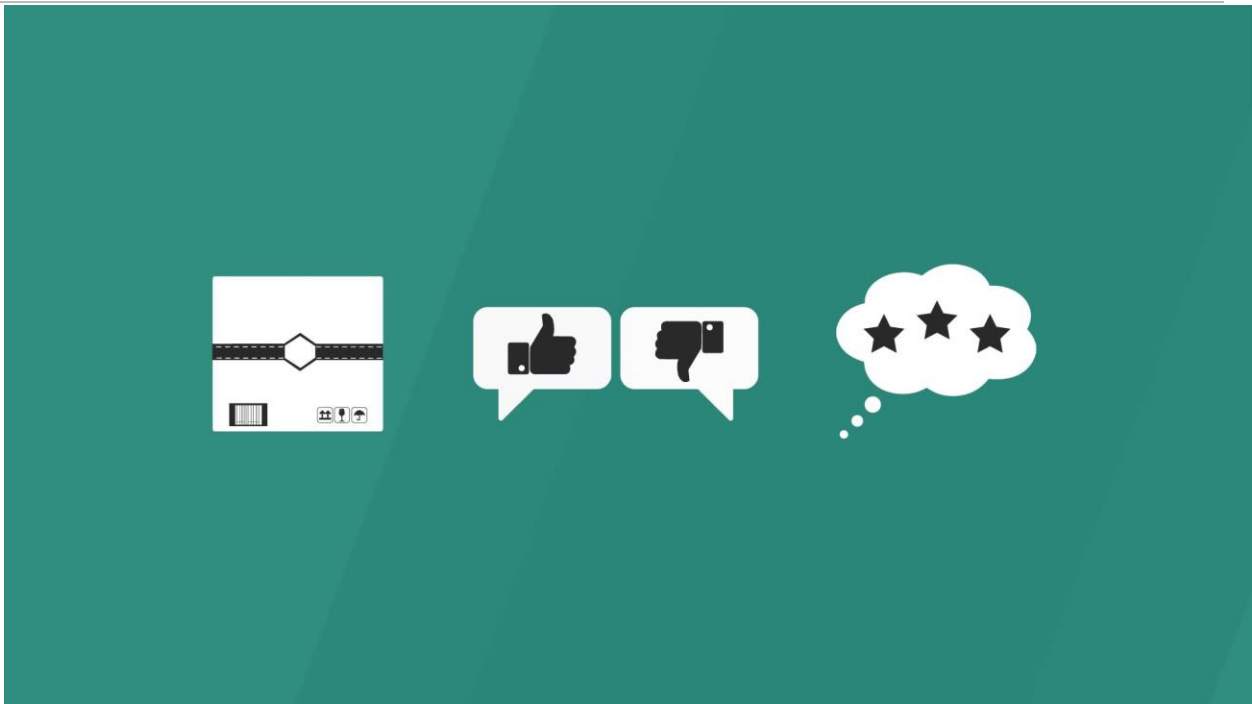
However, product managers have to consider other issues, such as cost, time, and other practical considerations, such as the availability of people and equipment to even conduct the experiment. As a product manager, you can use common methods to improve the reliability of your results. These are randomized field trials, blind tests, and data analysis. The first method, randomized field trials, involve using two similar sample groups. A change is applied to one group, the test group. And a control, or untested group, does not have the change. This enables you to compare the results between the test group and the control group. If the difference in results is significantly different, you can link that difference to the change in the test group. By using randomized testing, you can reduce systemic bias. In other words, you reduce the risk of people selecting samples that will give the results that they are hoping for. This in turn makes it easier to determine the cause of the different results between the two samples. The second method is blind tests. Experiments are susceptible to what's known as the Hawthorne Effect. This occurs when experiment participants alter the behavior, often without realizing it, because they know they are part of an experiment. In a blind test, participants are not aware that an experiment is taking place.

For example, a company might alter a component of a product without users being aware of the change. The third method to increase the reliability of experiment results is to use large volumes of data, and apply complex algorithms to analyze these multiple datasets. The results of this analysis will enable you to identify which changes have the most significant impact on the product's success. Your final question is: Will you get maximum value from the experiment? The costs involved in running experiments can be considerable, so you want to ensure that you get a good return on your investment of time and money.

In order to get a good return on investment, you need to understand if the experiment will provide insight into the product's impact on customers, markets, and

customer segments to suggest where the product is likely to find its greatest chance of success. By carefully analyzing and then exploiting the insights gained from your experiments, you can ensure that the experiments deliver value to the organization and the customers. Indeed, you should spend at least as much, if not more, time analyzing the data, and gathering key insights as you do collecting the data.

6. Video: The User-Experience-Based Design Approach (bs_apm01_a07_enus_06)



After completing this video, you will be able to identify the customer information collected by the user-experience-based design approach

Objectives

- *identify the customer information collected by the user-experience-based design approach*

[Video description begins] *Topic title: The User-Experience-Based Design Approach.*
[Video description ends]

As users think about or interact with a product, they experience responses, both positive and negative. These responses are referred to as the user experience, or UX. The International Organization for Standardization defines UX as, "persons' perceptions and responses resulting from the use and/or anticipated use of a product, system or service". By taking into account UX, product managers can design product experiments aimed at increasing users' positive experiences with the product, and decreasing their negative ones. By systematically capturing user responses, you can feed this information into future design decisions to drive better product delivery.

If a current product isn't meeting user's expectations, you need to find out why and rectify the situation. Ideally, user responses should be gathered as soon as possible in the planning and production phases, so that they can be considered as the product is developed. A key concept in user experience-based design of experiments, or UXDoE philosophy is that customer satisfaction plays a critical role in product quality. In other words, a quality product is one that satisfies customers and meets their expectations. There is sometimes a disconnect between what product designers think customers want, and what customers actually want. UXDoE enables you to carry out user research to bridge this gap.

You try to gather as much information about the customers as possible, including psychological, lifestyle, demographic, and technology factors. This information in turn drives product development effort. This is a very efficient approach to product development because it enables you to quickly define users' needs and expectations, and then build the product that will meet these needs. All product development takes place in the context of understanding what the user expects, thus reducing the chances of building a product that doesn't meet those expectations.

A typical UXDoE process involves three steps. First, you define the problem. Next, you identify user response and experimental factor and finally, you execute the experiment. Defining the problem involves establishing a link between the user's experience and the product design. Of course, you could collect data from almost limitless sources, with users having diverse backgrounds and experience. So, at this stage, you need to define which users you should focus on and then define the problem in terms of meeting their expectations. You also need a means of identifying and measuring user responses and experimental factors.

The experimental factor is the variable that you're focusing on, such as upload speed, visual appeal, or ease of interface navigation, for example. You might decide on using a rating system that users complete in order to record their satisfaction with different aesthetic elements of the product. Having defined the problem and how you're going to measure it, you can execute the experiment. This involves determining what form the experiment is going to take and who will participate in it. Product managers need to decide issues such as the scope of the experiment, the sequence of the tests, the management of the experiment environment, the collection of data, and the measurement of results. They may need to conduct some trial rounds before deciding on a final process. And once the experiment is up and running, it should be carefully monitored.

In order to understand user responses, product managers need to understand the users. They need to find out about their psychology and life factors. Are they comfortable using new products or do they resist change? Are they risk averse? Depending on the product, they may need to understand customer's attitudes

toward and acceptance of new technologies. Are they able to adapt to new technologies easily, or do they need a certain amount of support and coaching? Also, bear in mind their lifestyle and persona. Is the user a single mother on a tight budget who is looking for a bargain, or a retiree who likes to indulge in luxury products? And don't forget demographic variables. These include age, location, occupation, income, and so on. So how does UXDoE impact product design and development? First, it helps to clarify the conceptual activities that need to be carried out before building the product.

The customers' needs can drive the concepts that inform how the product is built. Once you have clarified the underlying concepts, you can begin teasing out the engineering details needed to turn the concepts into a concrete product that delivers what customers want. Using UXDoE, product designers can identify subjects who can take part in the experiments. These are the users who can provide the most valuable feedback about the product. UXDoE also helps designers decide which experimental factors to investigate. These are the product variables that can be measured to see their impacts on product performance and quality. And UXDoE enables the designers to then allocate the resources needed for the experiments. Designers and product managers can work together to decide budgets, team members, equipment, and other resources required. Next, you need to work with stakeholders to put in place the processes required to implement these engineering details. This involves building and checking the product, ensuring there are processes that enhance the quality of the finished product. In most cases, you'll start by manufacturing a prototype. This enables you to get further feedback before diving into full-scale production.

Of course, user feedback doesn't end with the launch of the product. Product managers must put in place post-launch activities that enable them to gather further information about the product. They can use this post-launch feedback to refine the product and address any quality issues that come to light. UXDoE benefits product managers by providing vital, early feedback about the product, its design, and usability. It allows product managers to evaluate the products, while they are still in the conceptual development and testing stages. Product managers also gain a deeper understanding of the products themselves.

By studying the user-product interaction, you can see which features are most popular with users. This helps determine what the key features are, which in turn can then drive not only the product design, but future marketing efforts. By getting a better understanding of which type of users are most positive about the product, product managers can identify target groups and develop marketing strategies around those audiences to maximize the product success in the marketplace.

7. Knowledge Check: Using Experimentation to Design Products

Product managers have many ideas on how to design their product to meet the customers' needs. The challenge is to determine which of those ideas have the best chance of working. Increasingly, product managers are using large-scale randomized experiments to test the effectiveness of their ideas.

In this course, you will learn how to use experimentation to make the best product management decisions, best practices for conducting effective experiments, the stages of conducting an experiment, how to optimize your product process with the design of experiments approach, and how to increase customer satisfaction with the user-experience-based design approach.

Objectives

- *discover the subject areas that will be covered in this course*
- *identify the reasons why product managers run experiments*
- *identify the steps for ensuring an effective experiment*
- *sequence the stages for conducting an experiment*
- *identify the questions to ask to see if an experiment is worth the expense and effort*
- *identify the customer information collected by the user-experience-based design approach*

Question 1: Multiple Choice

Why should product managers run experiments on ideas and products?

Options:

1.
To develop new ideas and concepts
2.
To educate customers about the product
3.
To improve the product manager's career path
4.
To improve product reliability
5.
To test materials, processes, and devices
6.
To optimize product performance

Answer

1.
To develop new ideas and concepts
4.
To improve product reliability

5.

To test materials, processes, and devices

6.

To optimize product performance

Feedback:

Option 1:

This option is correct. Product managers should focus on testable ideas that can be scientifically evaluated.

Option 2:

This option is incorrect. The aim of experimenting is to better understand what customers want, rather than to educate them.

Option 3:

This option is incorrect. The focus of experimenting should be on the product, not the product manager's career.

Option 4:

This option is correct. By implementing the findings from experiments, product managers can make products more reliable.

Option 5:

This option is correct. Experimenting enables product managers to test materials, processes, and devices that are used to develop new products.

Option 6:

This option is correct. Experimenting can unearth flaws in a product that can impact its performance.

Question 2: Multiple Choice

What steps are involved in ensuring an effective experiment?

Options:

1.

Ignoring unexpected results

2.

Paying the participants

3.

Reporting your errors

4.

Tracking info by recording data

5.

Drawing your conclusion from results

6.

Using control

Answer

3.

Reporting your errors

4.

Tracking info by recording data

5.

Drawing your conclusion from results

6.

Using control

Feedback:

Option 1:

This option is incorrect. The aim of experimenting is to find data that either proves or disproves an assumption. You cannot ignore results just because you don't like them.

Option 2:

This option is incorrect. Paying participants might affect how they behave in the experiment, which in turn might skew the results.

Option 3:

This option is correct. Errors are almost inevitable in any experiment. It's important to keep track of these errors and learn from them.

Option 4:

This option is correct. Experimenting involves collecting and analyzing measurable data about the idea or product.

Option 5:

This option is correct. Experiments yield data from which you should be able to draw logical conclusions.

Option 6:

This option is correct. An experiment usually involves a control scenario and an experimental scenario. The control scenario reflects the current situation, while the experimental scenario applies changes to the current situation.

Question 3: Ranking

Put the stages for conducting an experiment in order.

Options:

A.

Communicate your findings and close the loops

B.

Ask questions and generate ideas

C.

Conduct your experiment

D.

Create a hypothesis

Answer

1.

Ask questions and generate ideas

The starting point of an experiment is a testable idea. Asking questions will enable you to brainstorm and generate product ideas.

2.

Create a hypothesis

The second step is to create a hypothesis. You need to develop your testable idea into a hypothesis that can be either proved or disproved. Most hypotheses take the form of "if...then" statements.

3.

Conduct your experiment

Step three is to conduct your experiment. The aim of the experiment is to test the hypothesis. Common methods include A/B testing and multivariate testing.

4.

Communicate your findings and close the loops

Communicating findings is the fourth and final step. When the experiment is completed, you should communicate the findings in an easy-to-understand format and clearly outline what actions need to be taken.

Question 4: Multiple Choice

Which questions should you ask to determine if an experiment is worth the expense and effort?

Options:

1.

Is the experiment doable?

2.

Can you ensure reliable results?

3.

Will you get maximum value from the experiment?

4.

Have competitors developed a similar product?

5.

Have stakeholders made a commitment to abide by the results?

6.

How will you communicate the product changes to your customers?

7.

Does the experiment have a clear purpose?

Answer

1.

Is the experiment doable?

2.

Can you ensure reliable results?

3.

Will you get maximum value from the experiment?

5.

Have stakeholders made a commitment to abide by the results?

7.

Does the experiment have a clear purpose?

Feedback:

Option 1:

This option is correct. In a complex project, there may be many variables, and many interactions between them.

Option 2:

This option is correct. You can use randomized field trials, blind tests, and big data to improve the reliability of your results.

Option 3:

This option is correct. Experiments cost time and money, so you need to ensure there will be a good return on investment.

Option 4:

This option is incorrect. When conducting experiments, you need to focus on your own products, not competitors' products.

Option 5:

This option is correct. You need to get agreement from stakeholders regarding how they'll act once the experiment results are clear. They cannot simply pick the data that confirms their own biases.

Option 6:

This option is incorrect. This is a question for the Marketing Department to address after the experiment's results have been implemented.

Option 7:

This option is correct. You need to consider whether an experiment is the best practical way to solve the business problem. Think about what you need to learn to answer the question.

Question 5: Multiple Choice

Which types of customer information are usually collected when using a user-experience-based design approach?

Options:

1.
Psychology
2.
Health history
3.
Science and technology acceptance
4.
Life factors
5.
Appearance
6.
Demographic variables
7.
Lifestyle
8.
Persona

Answer

1.
Psychology
3.
Science and technology acceptance
4.
Life factors
6.
Demographic variables
7.
Lifestyle
8.
Persona

Feedback:

Option 1:

This option is correct. It's useful to get a sense of the users' psychology. Do they tend to be loyal, long-term users, or do they like to constantly try new gadgets?

Option 2:

This option is incorrect. Although health history would be useful data, you would not be able to access this information.

Option 3:

This option is correct. Some customers are keen to embrace new technologies, whereas others are more comfortable with existing technology.

Option 4:

This option is correct. Life factors includes a person's well-being and environment.

Option 5:

This option is incorrect. A user's appearance would not have an impact on their experience of using the product.

Option 6:

This option is correct. Demographic factors include things like age, race, relationship status, address, and so on.

Option 7:

This option is correct. Understanding a user's lifestyle will enable you to develop products for them. For example, do they prefer luxury goods or a bargain? Do they have energetic or relaxing pastimes?

Option 8:

This option is correct. A persona enables you to create an imaginary holistic picture of a user.

Course HTML Resources

- Glossary: Product Management: Designing and Running Experiments

business case	A proposal for a new product or service that will benefit the business.
cause and effect	A relationship between two variables where a change in one variable directly causes a change in the other. <i>See also variable.</i>
input factor	An item or resource in a process that produces an output.

metric	A quantified measurement used to evaluate results.
minimum value product	A simple version of a product that delivers the minimum amount of value that the customer expects.
objectives and key results	<i>See OKR.</i>
OKR	Abbreviation for objectives and key results, the key metrics used to evaluate the performance of a product. <i>See also metrics.</i>
proof of concept	An assessment of the viability of a business idea.
prototype	A simple, experimental version of a product to test its basic functionality and viability.
scientific method	A systematic, logical approach to conducting an experiment and evaluating its results. It involves working from a hypothesis and then gathering the necessary data to test that hypothesis and draw logical conclusions.
segmentation	Breaking up, or segmenting, a large data set into smaller, logically grouped sets to get a deeper understanding of what the data is revealing.
user persona	A description of an ideal or imagined user. It could include demographic details such as age and location, as well as technological expertise, values, and outlook.
variable	A value that changes, or varies, depending on circumstances.

Question 1 of 5

Question: Which of these steps are used when conducting experiments?

Instruction: Choose all options that best answer the question.

- Tracking info by recording data
- Using control
- Drawing a conclusion from the results
- Reporting errors
- Continuing to repeat the experiment
- Using as small a team as possible

Question 4 of 5

Question: What kind of customer information is collected by the user-experience-based design approach?

Instruction: Choose all options that best answer the question.

- Lifestyle
- Psychology
- Science and technology acceptance
- Phone number
- Size
- Demographic variables

Question 5 of 5

Question: What are common reasons for product managers to conduct product experiments?

Instruction: Choose all options that best answer the question.

- To test materials, processes, and devices
- To upskill the production team
- To improve product reliability
- To optimize product performance
- To develop new ideas and concepts
- To increase the price of products

Question: Why should product managers run experiments on ideas and products?

Instruction: Choose all options that best answer the question.

- To educate customers about the product
- To develop new ideas and concepts
- To improve the product manager's career path
- To test materials, processes, and devices
- To optimize product performance
- To improve product reliability

Question 2 of 5

Question: What steps are involved in ensuring an effective experiment?

Instruction: Choose all options that best answer the question.

- Tracking info by recording data
- Paying the participants
- Drawing your conclusion from results
- Reporting your errors
- Using control
- Ignoring unexpected results